

PRODIPTA SEN AMARTYA

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EDUCATION

University of Dhaka • Dhaka, Bangladesh January 2019 – December 2023

Bachelor of Science • *Computer Science and Engineering* • CGPA: 3.49/4.0

• *Academic Transcripts*

Test of English as a Foreign Language (TOEFL iBT) • November 2024

Overall: 95 • Reading: 25 • Listening: 22 • Speaking: 24 • Writing: 24

RESEARCH INTERESTS

- AI for Security; Trustworthy and Interpretable AI; Computer Security; Software Security; Decision-making under Uncertainty

PUBLICATIONS

- **Prodipta Sen Amartya**, Shaily Kabir, Sagar Chandra Karmaker Babu, and Mosarrat Jahan. “An Interval Creation Approach to Construct Interval Type-2 Fuzzy Sets.” *2024 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, IEEE World Congress on Computational Intelligence (WCCI), 2024. ([Paper](#) | [Slides](#))
- Tahmid Imtiaz Imu, Emama Nahid, **Prodipta Sen Amartya**, et al. [Title withheld pending review]. **Under review**, *36th USENIX Security Symposium (USENIX Security '27)*, Cycle 1.
- **Prodipta Sen Amartya**, Tahmid Imtiaz Imu, Shaily Kabir, and Christian Wagner. “Towards Improved Movie Recommendations via Handling of Input Uncertainty.” **Working paper**, 2025, LUCID Lab, University of Nottingham, United Kingdom. ([Paper](#))

RESEARCH EXPERIENCE

Backdoor Attacks in Large Language Model Reasoning

Supervisor: Dr. Honghui Xu,

Kennesaw State University, Georgia, USA

July 2026 – Present

- Investigating vulnerabilities in large language models’ chain-of-thought reasoning under adversarial backdoor triggers
- Designing and evaluating a novel backdoor attack strategy targeting multi-step reasoning; manuscript currently under review at *USENIX Security '27*
- **Technical Skills:** LLM reasoning evaluation (chain-of-thought analysis), backdoor/trigger injection techniques, Python, PyTorch, Hugging Face Transformers

Security Analysis of Federated Learning and Split Learning

Supervisor: Dr. Honghui Xu,

Kennesaw State University, Georgia, USA

July 2026 – Present

- Systematizing attack patterns and defense mechanisms across federated learning and split learning paradigms
- Evaluating the generalizability of existing defenses when applied across federated and split learning settings

- Establishing theoretical grounds and evaluation metrics for the transferability of attack and defense methods between the two paradigms
- Co-authoring a Systematization of Knowledge (SoK) paper targeting *IEEE S&P (Oakland) 2027*
- **Technical Skills:** CNN, GAN, adversarial attacks (model inversion, membership inference, gradient leakage), defense mechanisms (differential privacy, secure aggregation), Explainable AI

Improving Security of Systems Integrated in Ultrasound Devices

Samsung R&D Institute Bangladesh, Dhaka, Bangladesh

August 2026 – Present

- Investigating how updates to independently developed ultrasound software modules — DICOM server, rendering engine, authentication, device drivers — can silently invalidate the security assumptions neighboring modules rely on
- Developing a contract-based method to model those assumptions and scope re-verification to the modules a change can affect, rather than retesting the full stack
- Designing automated checks that run at deployment time, surfacing cross-module security regressions that per-module testing does not catch
- **Technical Skills:** CodeQL, Semgrep, pydicom, NetworkX, Syft/CycloneDX

Sensitivity Analysis with Fuzzy Sets

Supervisor: Dr. Shaily Kabir

LUCID Lab, University of Nottingham, UK

October 2024 – March 2025

- Identified which input variables and fuzzy rules govern the robustness of a fuzzy inference system under increasing input uncertainty
- Built a non-singleton fuzzy logic (NSFLS) recommender over MovieLens, outperforming collaborative filtering and type-1 fuzzy baselines as injected noise increased
- First-authored the white paper “Towards Improved Movie Recommendations via Handling of Input Uncertainty” ([Paper](#))
- **Technical Skills:** FuzzyR, non-singleton fuzzy logic systems, explainable AI

Evaluation of the Behavior and Performance of Interval Type-2 Fuzzy Sets

University of Dhaka, Dhaka, Bangladesh

March 2023 – January 2024

- Conducted a comparative efficiency analysis across type-1, general type-2, and interval type-2 fuzzy sets to characterize where the added expressiveness of higher-order sets justifies its computational cost
- Designed the Interval Creation Approach (ICA), a methodology constructing interval type-2 fuzzy sets from type-1 sets of arbitrary shape — triangular, trapezoidal, Gaussian, generalized bell-shaped, symmetric, and asymmetric
- Benchmarked ICA on synthetic and real-world datasets against the Interval Approach (IA), Enhanced Interval Approach (EIA), and M2, reducing pre-processing overhead and overall computational cost
- First author of “[An Interval Creation Approach to Construct Interval Type-2 Fuzzy Sets](#)”, published from this research
- **Technical Skills:** Python, MATLAB, PYIT2FLS, Juzzy

ACADEMIC SERVICE

Reviewer

Workshop on Trustworthy AI for Good (AI4GOOD), ICML

2026

- Reviewed submissions for the AI4GOOD workshop at the *International Conference on Machine Learning* ([Workshop](#) | [Certificate](#))

Judge and Question Setter

Bangladesh Artificial Intelligence Olympiad

2026

- Set problems and evaluated submissions for the regional stage, contributing to candidate selection for the national round

INDUSTRY EXPERIENCE

Senior Software Engineer

Samsung R&D Institute Bangladesh, Dhaka, Bangladesh

March 2026 – Present

■ *Project: Cardiac TEE Rendering – 5D Viewer, Samsung Medison*

A rendering platform for interactive visualization of 3D/4D cardiac ultrasound data.

- Resolved real-time sync failures in the clinical UI by redesigning cross-process polling, eliminating race conditions in viewer-engine RPC
- Designed the interaction layer for rendered volumes: rotation, arbitrary-plane slicing, repositioning
- Implemented the .NET-to-C++ pipeline binding the viewer UI to the rendering engine
- Refactored the UI codebase to reduce interaction latency under continuous rendering
- Extended label rendering to full Unicode coverage across supported locales
- **Technical Skills:** C++, C#, .NET, HLSL, DirectX

Software Engineer

Samsung R&D Institute Bangladesh, Dhaka, Bangladesh

March 2024 – February 2026

■ *Project: Cardiovascular Workstation – HelloMom, Samsung Medison*

Cloud and on-premises DICOM server for Samsung ultrasound machines, accessed through paired Windows and web clients.

- Built authentication end-to-end: login, sign-up and single sign-on across Samsung Medison applications, incorporating Google OAuth 2.0
- Designed and implemented role-based access control governing file permissions across the system
- Developed the DICOM tag parser organizing studies into a structured server-side hierarchy, and the REST APIs serving both clients
- Implemented the full UI component set for the WPF desktop and React web clients, maintaining behavioral parity across platforms
- Deployed and configured the on-premises hosting environment on IIS/Windows Server for organization-wide access
- **Technical Skills:** C#, .NET, WPF, React, Next.js, Redux Toolkit, OAuth 2.0, DICOM, IIS Windows Server

■ *Project: Cloud Solutions – SonoSync, Samsung Medison*

Real-time streaming and remote operation of ultrasound machines over the web.

- Owned real-time publication of the ultrasound stream to web clients
- Replaced the project's ffmpeg dependency with a custom stream reader, implementing double buffering over DirectShow and Media Foundation for direct frame-pipeline control
- Built the cloud integration enabling remote control of machine components
- Contributed to the real-time transport layer between machine and remote clients
- **Technical Skills:** C++, .NET, DirectShow, Media Foundation, WebRTC

ACADEMIC PROJECTS

Gardening Tukitaki

University of Dhaka, Dhaka, Bangladesh

2023

- Built a full-stack web platform combining a community blog with a marketplace for seeds and agricultural products, serving gardeners, agriculturists, and farmers ([Project Code](#))
- Designed the relational schema and backend APIs serving both, including a fertilizer calculator deriving input quantities for a crop from published agricultural studies
- **Technical Skills:** Python, Django, SQLite3, JavaScript, HTML/CSS

Find Your Doctor

University of Dhaka, Dhaka, Bangladesh

2022

- Developed a native Android application for discovering practitioners and booking appointments, with distinct role models for patients and practitioners managing their own availability ([Project Code](#) | [Documentation](#))
- Implemented the backend and interface design, and led the functional and non-functional requirements analysis
- **Technical Skills:** Java, Firebase, Android SDK, automated testing

Book Your Ride

University of Dhaka, Dhaka, Bangladesh

2021

- Engineered an Android intercity bus ticketing system with schedule lookup by route and date, issuing digital tickets in place of physical counter sales ([Project Code](#))
- Designed the backend architecture and seat inventory model, resolving concurrent booking requests for the same seat through Firebase transactions to prevent double-allocation
- **Technical Skills:** Java, Firebase, Android SDK

Library Management System

University of Dhaka, Dhaka, Bangladesh

2020

- Designed a normalized relational database for an institutional library, enforcing referential integrity across circulation records through foreign key constraints ([Project Report](#))
- Implemented views abstracting the recurring queries behind routine library operations
- **Technical Skills:** SQL, Oracle Database

AWARDS & HONORS

- 1st position out of 400 participants, Basic AI/ML Exam, 2025, Samsung R&D Institute Bangladesh
- Talent Pool Scholarship in HSC, 2019-2023, Dhaka, Bangladesh
- Agriculturist (Krishibid) Institute Scholarship, 2019-2023, Dhaka, Bangladesh

CO-CURRICULAR ACTIVITIES

Online courses -

- Generative AI & Deep Learning training | BUET × SRBD | [Certificate](#)
- Machine Learning & Data Science: The Complete Visual Guide | Udemy | [Certificate](#)
- Microservices: Designing Highly Scalable Systems | Udemy | [Certificate](#)
- Website Hacking / Penetration Testing | Udemy | [Certificate](#)

EXTRA-CURRICULAR ACTIVITIES

- Co-organizer, Study Group on Vulnerabilities of Self-Adapting Large Language Models, 2026 – Present, East West University, Dhaka, Bangladesh
- Volunteer, Code Samurai by BJIT, 2022, University of Dhaka, Dhaka, Bangladesh

SKILLS

- **Programming Languages:** C, C++, C#, Java, Python, JavaScript/TypeScript, SQL
- **Databases:** SQLite3, Firebase Real-time Database & Cloud Firestore
- **Framework & SDK:** .NET, Django, Android SDK, Bootstrap, React, Next.js, Node.js
- **Operating System:** Windows, Linux (Debian)
- **Machine Learning Tools:** NumPy, Pandas, scikit-learn
- **Web Languages:** HTML, CSS
- **Security & ML Research:** PyTorch, Hugging Face Transformers, CodeQL, Semgrep, FuzzyR
- **Other Tools:** Docker, Azure, WebRTC, CI/CD (GitHub Actions), REST APIs

REFERENCES

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| <ul style="list-style-type: none">• Dr. Honghui Xu
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